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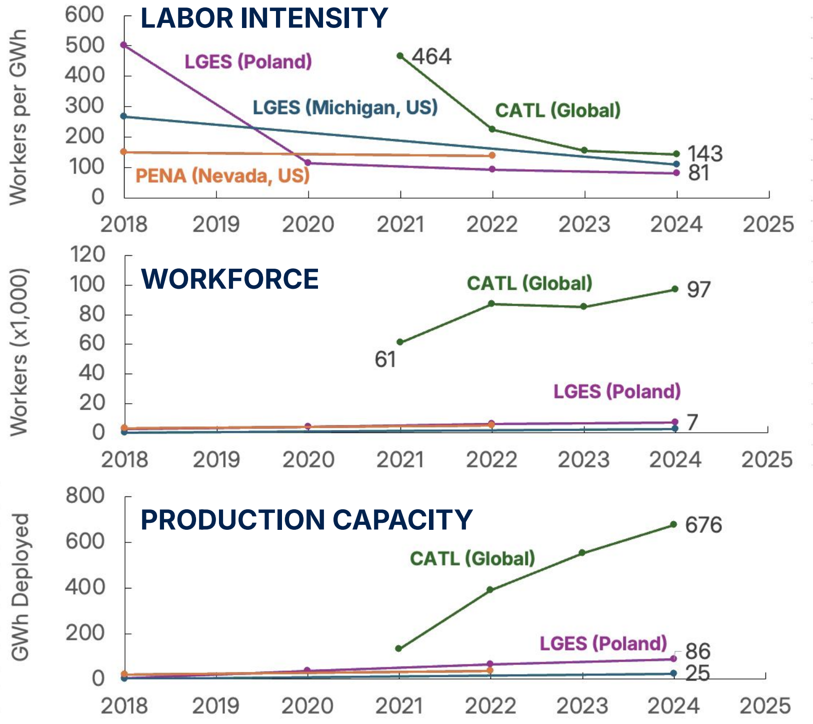
How Many Workers Are Needed for Cell Manufacturing?

Battery cell manufacturing **labor intensity peaks during production ramp-up, often exceeding 400 workers per GWh.**

As production capacity scales, process automation and process maturity improves efficiency, requiring fewer workers per GWh.

At steady-state production, **the labor intensity declines sharply**, with **best-in-class plants approaching 80 workers per GWh**:

- CATL (Global): 143 workers/GWh[1]
- LGES (Michigan, US): 108 workers/GWh[2,3]
- PENA (Nevada, US): 137 workers/GWh[4,5]
- LGES (Wrocław, Poland): 81 workers/GWh[6]



Data summarizes publicly-available employment and factory production data from company reports, press releases, and news reports. The data was compiled and summarized by Krish Lulla (krishisd@umich.edu) as part of an independent research project at the University of Michigan. Production capacity corresponds to realized or deployed capacity, not promised capacity.

Sources: 1. CATL, 2. LG Energy Solution, 3. Battery Technology Online, 4. NAATBatt, 5. Automotive Manufacturing Solutions, 6. LG Energy Solution.